AIAC LAB 2

AI-Assisted Coding – Responsible AI

Task 1 – Privacy & Security in File Handling

Question: Use an AI tool to generate a Python script that stores user data (name, email, password) in a file. Analyze whether sensitive data is stored in plain text or without encryption.

Python Code (VS Code screenshot)

VS Code • lab2_task1.py Python

```python
import hashlib

name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

hashed_password = hashlib.sha256(
    password.encode()
).hexdigest()

with open("user_data.txt", "w") as file:
    file.write(f"Name: {name}\n")
    file.write(f"Email: {email}\n")
    file.write(f"Password Hash: {hashed_password}\n")

print("User data saved securely.")
print("Password is stored as a hash.")
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task1.py
Enter name: Lakshitha
Enter email: lakshitha@gmail.com
Enter password: abc123

User data saved securely.
Password is stored as a hash.
```

Task 2 – Fair Decision Logic

Question: Generate a scholarship eligibility checker based on academic score, family income, and location. Examine whether the logic unfairly favors or disadvantages any group.

Python Code (VS Code screenshot)

VS Code • lab2_task2.py Python

```python
academic_score = float(input("Enter academic score: "))
income = float(input("Enter family income: "))
location = input("Enter location: ")

eligible = (
    academic_score >= 75
    and income <= 300000
    and location.lower() in ["rural", "urban"]
)

if eligible:
    print("Scholarship Eligible")
else:
    print("Scholarship Not Eligible")

print("Decision is based on score and income.")
print("Location is not used to favor a group.")
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task2.py
Enter academic score: 82
Enter family income: 250000
Enter location: rural

Scholarship Eligible
Decision is based on score and income.
Location is not used to favor a group.
```

Task 3 – Transparency in AI-Based Admission System

Question: Generate a Python program that predicts whether a student should be selected or rejected for admission using entrance exam score and academic marks. The system must provide a clear reason for its decision.

Python Code (VS Code screenshot)

VS Code • lab2_task3.py Python

```python
entrance = float(input("Enter entrance exam score: "))
marks = float(input("Enter academic marks: "))

score = (entrance * 0.6) + (marks * 0.4)

if score >= 70:
    decision = "Selected"
    reason = "The combined score meets the admission threshold."
else:
    decision = "Rejected"
    reason = "The combined score is below the admission threshold."

print("Admission Decision:", decision)
print("Reason:", reason)
print("Combined Score:", round(score, 2))
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task3.py
Enter entrance exam score: 78
Enter academic marks: 82

Admission Decision: Selected
Reason: The combined score meets the admission threshold.
Combined Score: 79.6
```

Task 4 – Ethical Scoring System

Question: Generate an employee performance evaluation system using project completion rate, teamwork score, and attendance. Analyze the scoring logic for unethical weighting or hidden bias.

Python Code (VS Code screenshot)

VS Code • lab2_task4.py Python

```python
completion = float(input("Project completion rate (%): "))
teamwork = float(input("Teamwork score (%): "))
attendance = float(input("Attendance (%): "))

performance = (
    completion * 0.50
    + teamwork * 0.30
    + attendance * 0.20
)

print("Performance Score:", round(performance, 2))

if performance >= 75:
    print("Evaluation: Good Performance")
else:
    print("Evaluation: Needs Improvement")

print("Weights are based on job-related criteria.")
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task4.py
Project completion rate (%): 90
Teamwork score (%): 80
Attendance (%): 95

Performance Score: 87.0
Evaluation: Good Performance
Weights are based on job-related criteria.
```

Task 5 – Accessibility and Inclusiveness

Question: Generate a user feedback form application. Review whether the language and options are inclusive and accessible to diverse users.

Python Code (VS Code screenshot)

VS Code • lab2_task5.py Python

```python
print("=== User Feedback Form ===")

name = input("Name (optional): ")
feedback = input("Please share your feedback: ")

print("\nHow was your experience?")
print("1. Very Good")
print("2. Good")
print("3. Neutral")
print("4. Needs Improvement")
rating = input("Choose an option (1-4): ")

if rating in ["1", "2", "3", "4"]:
    print("\nThank you for your feedback!")
    print("Your response has been recorded.")
else:
    print("\nPlease choose an option from 1 to 4.")

print("Name is optional and the form uses clear language.")
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task5.py
=== User Feedback Form ===
Name (optional): Lakshitha
Please share your feedback: Easy to use

How was your experience?
1. Very Good
2. Good
3. Neutral
4. Needs Improvement
Choose an option (1-4): 1

Thank you for your feedback!
Your response has been recorded.
Name is optional and the form uses clear language.
```

Task 6 – Fairness in Loan Approval System

Question: Generate a Python program that predicts whether a person should be approved or rejected for a loan based on income, age, employment status, credit score, and loan amount. Ensure the decision does not use sensitive attributes such as gender, religion, or race.

Python Code (VS Code screenshot)

VS Code • lab2_task6.py Python

```python
income = float(input("Enter annual income: "))
age = int(input("Enter age: "))
employment = input("Employment status: ")
credit_score = int(input("Enter credit score: "))
loan_amount = float(input("Enter loan amount: "))

approved = (
    income >= 300000
    and age >= 18
    and employment.lower() == "employed"
    and credit_score >= 650
    and loan_amount <= income * 0.5
)

if approved:
    print("Loan Decision: Approved")
    print("Reason: Financial criteria are satisfied.")
else:
    print("Loan Decision: Rejected")
    print("Reason: One or more financial criteria are not satisfied.")

print("No gender, religion, or race is used.")
```

Output (VS Code Terminal screenshot)

VS Code • Terminal

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task6.py
Enter annual income: 600000
Enter age: 25
Employment status: employed
Enter credit score: 720
Enter loan amount: 200000

Loan Decision: Approved
Reason: Financial criteria are satisfied.
No gender, religion, or race is used.
```